

as the approaches above. But it does provide guidelines to adapt to changing liquidity conditions (e.g., transact more shares in times of higher market volumes and transact less shares in times of lower market volume) throughout the trade periods which is not provided from any of the previously described techniques. And as we show below, it provides the quickest solutions even for large trade lists.

Comparison of Optimization Techniques

To compare the performance of the different optimization techniques we conducted a simulation experiment to measure solution time and accuracy. The experiment is as follows:

Sample Universe. Our sample universe was the SP500 index as of December 31, 2011.

Number of Stocks. We constructed portfolios that ranged in size from 10, 25, 50, 100, . . . 450, and 500 stocks.

Order Size. We randomly defined order sizes from 0–25% ADV.

Volatility. We used actual stock volatility from the sample.

Covariance Matrix. We constructed our covariance matrix using a correlation between stocks that was equal to the average stock sector to sector correlation. For example, if the average correlation between a technology and utility stock was $\rho = 0.15$ we used a correlation of 0.15 to compute the covariance between a technology stock and a utility stock along with their actual volatility.

Number of Simulations. We performed twenty simulations for each portfolio for each optimization technique. Ten simulations were performed for a one-sided portfolio, e.g., cash investment, five simulations were performed using a two-sided portfolio with equal weights in each side, and five simulations were performed using a 130-30 two-sided portfolio, that is, the dollar weights in one side were 130% of the total and the weights in the other side accounted for -30% of the dollar value.

Performance Measure. We recorded the time to solve each portfolio with each optimization technique and also measured the accuracy of each technique by comparing the resulting trade schedule to the true trade schedule determined by solving the portfolio using the non-linear optimization routine that solved the exact objective formulation (Equation 9.5). Advantages and disadvantages of each technique are shown in Table 9.2.